Shaif Ahamed *Tamim.*

I build intelligent AI systems with a purpose.



AI ENGINEER & RESEARCHER · AGENTIC AI · COMPUTER VISION · AWS

MSc in Computer Vision, Robotics and Machine Learning — University of Surrey

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LINKEDIN	linkedin.com/in/shaif111	GITHUB	github.com/shaif143
SCHOLAR	Google Scholar profile	LIVE	ClinicalOKF · ClinicalAI

3 years hands-on AI / ML & research

2 live AI agents clinical assistants on Vercel

6 research papers in healthcare AI

3 research teams mentored at AIRIL

01 PROFILE

AI engineer and researcher with **around three years of hands-on experience** taking ML systems from idea to production — document-intelligence pipelines, deep-learning models for healthcare and, most recently, **multi-agent systems built with LangGraph, n8n and open-weight Hugging Face models**, grounded in Google's Open Knowledge Format and deployed on AWS. I combine a **PyTorch-first research background** with full-stack delivery skills, so I can design the model *and* ship the product around it. I am studying for an **MSc in Computer Vision, Robotics and Machine Learning at the University of Surrey**, after completing IIT Roorkee's Advanced PG Certificate in AI Engineering on Cloud and AIOps. Known for clear communication and an approachable, collaborative style.

> **Availability:** part-time & remote AI/ML roles alongside the MSc · full-time from September 2027 · open to relocating.

02 TECHNICAL SKILLS

bold = tools I use most

LANGUAGES & FRAMEWORKS	**Python (advanced)**, **TypeScript / JavaScript**, **Next.js**, **NestJS**, C++, SQL
ML & DEEP LEARNING	**PyTorch (expert)**, **TensorFlow / Keras**, **Scikit-learn**, Pandas, NumPy · CNN, ANN, LSTM, Transformers, BERT, transfer learning · Random Forest & ensembles · explainable AI (XAI) · anomaly detection · Bayesian tuning · evaluation & cross-validation · reinforcement learning
COMPUTER VISION & EDGE	**OpenCV**, **YOLO** · image & video data · sensor-signal processing · Raspberry Pi, Arduino · ROS 2 (currently learning in MSc)
GENERATIVE AI & LLMS	**Hugging Face (open-weight models)**, **RAG**, OpenAI API, Gemini API · prompt engineering · fine-tuning · embeddings & sentence-transformers · vector databases (FAISS, Pinecone, Weaviate, Chroma) · chunking strategies · GANs, VAEs, diffusion · responsible AI (hallucination, drift, bias)
AGENTIC AI & KNOWLEDGE	**LangGraph**, **LangChain**, **n8n**, **Google Open Knowledge Format (OKF)** as a governed knowledge layer for agents · AutoGen, SmolAgents, LlamaIndex · router & specialist agents · tool calling · hierarchical planning · memory & retrieval augmentation · human-in-the-loop
NLP & DOCUMENTS	TF-IDF, BoW, Word2Vec · text classification · information extraction · summarisation · document intelligence (PDF / DOCX / CSV) · multilingual retrieval (Bangla + English)
CLOUD, MLOPS & DEVOPS	**AWS** (EC2, Lambda, S3, API Gateway, IAM, Auto Scaling, Bedrock, SageMaker) · **Docker** · Kubernetes & Istio · CI/CD for AI APIs · model versioning & rollbacks · monitoring & logging · A/B testing · GPU workloads · cost optimisation · zero-trust security · Linux, Git
DATA & DATABASES	**PostgreSQL**, **MongoDB**, SQL / NoSQL · EDA & feature engineering · data quality & versioning · statistical, image, video, audio, text and sensor time-series data
WEB & MOBILE	**React & React Native**, Node.js, FastAPI · REST API design · Vercel · IEEE 830 SRS & technical documentation · data structures & algorithms · human–computer interaction
SOFT SKILLS	Leadership & team management · mentoring & coaching · cross-functional collaboration · stakeholder communication · presentation & technical articulation · prioritisation & time management · delivering under strict deadlines · constructive feedback · high-productivity team culture · results-driven · detail-oriented · problem solving · adaptability

03 PROFESSIONAL EXPERIENCE

AI & App Developer *HyperGlow UK*
Jun 2026 – Aug 2026

Leicester, UK · AI-powered content for LED/LCD video walls, digital signage and hospitality ordering

- Built an **ML model of customer ordering patterns** (weather, day, date, mood) that powers customer segmentation and engagement features used in business decisions.
- Developed an **AI image-enhancement app** for restaurant food photography used in menus and digital signage.
- Designed and shipped a **restaurant self-service ordering app with behavioural upsell** (Next.js, PostgreSQL, Docker, React Native), deployed on a Linux server.
- Turned requirements from operations, creative and client stakeholders into technical specifications; wrote an **IEEE 830-style SRS**, handover documentation, an investor pitch deck and a financial model.

Research Assistant & Research Team Lead *AIRIL — AI Research & Innovation Lab*
Jan 2024 – Aug 2026

Dhaka, Bangladesh

- Led and **mentored three research teams** in healthcare AI, explainable AI, NLP and deep learning: reviewed experiments, code and paper drafts, and gave constructive feedback.
- Built and tuned **deep-learning pipelines in PyTorch and TensorFlow** (CNN, ANN, BERT, ensembles, XAI) for cancer classification, human activity recognition and disease diagnosis.
- Ran statistical experiments end to end — Pandas data cleaning, feature engineering, training, hyperparameter tuning and evaluation against baselines.
- Contributed to **six research papers**; maintained performance-tracking dashboards and cross-team reporting.

AI/ML Researcher & Software Engineer (Intern) *PAP International Ltd.*
Aug 2023 – Dec 2023

Mohakhali DOHS, Dhaka, Bangladesh

- Designed and deployed **production-ready NLP pipelines** for text classification, information extraction and document intelligence using Python, LangChain and the OpenAI API.
- Built **multi-source data-ingestion pipelines** (REST APIs, SQL/NoSQL databases, PDF/DOCX/CSV files) with data-quality controls and versioning.
- Developed scalable **REST APIs** that expose AI capabilities to front-end apps; selected and integrated LLMs on **cost, performance and security** criteria, delivering under strict deadlines.

04 EDUCATION & ADVANCED TRAINING

MSc in Computer Vision, Robotics and Machine Learning
University of Surrey

Guildford, UK · Sep 2026 – Sep 2027 (expected)

Advanced PG Certificate in AI Engineering on Cloud and AIOps
Continuing Education Centre, IIT Roorkee

with Futurense Technologies · Sep 2025 – Jun 2026 · Completed · ID CA-17-2025-26/78

- **Coursework:** ML & deep learning (CNN, YOLO, LSTM) · GenAI & fine-tuning · RAG · agentic AI (LangGraph, AutoGen) · AWS (Bedrock, SageMaker), Kubernetes, CI/CD, MLOps.

BSc in Computer Science & Engineering
American International University-Bangladesh (AIUB)

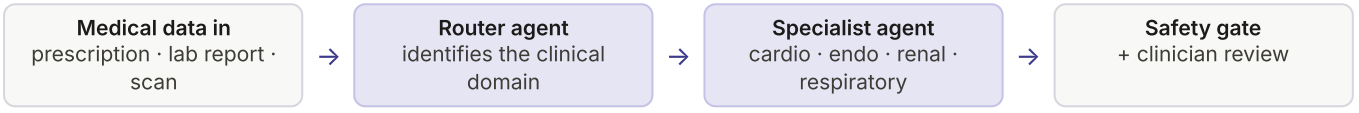
Dean's Honourable Mention List ×2 · final-year thesis on deep learning for healthcare

HSC & SSC (Science)
Govt. Science College, Dhaka

SSC — Rajendrapur Cantonment Public School & College

05 SELECTED PROJECTS

MediRoute AI *multi-agent clinical routing*
IN DEVELOPMENT



- A router agent reads prescriptions, lab reports and scans, identifies the clinical domain and **hands off to a specialist agent** trained on it.
- **Open-weight Hugging Face models** grounded in OKF — no paid LLM API; **LangGraph + n8n**, Docker, architected for **AWS** (Lambda, SageMaker, S3).

ClinicalOKF — LIVE

clinical-assistant-okf.vercel.app · independent project

- **Governed clinical ontology in Google's OKF v0.2** instead of vector search — 12 concepts, 17 sources, **zero vector DBs**.
- **Privacy-aware router** (local engine or Gemini Flash), safety gates, audit trail.

Stack OKF v0.2 · rule engine · Gemini Flash · Vercel

ClinicalAI — LIVE

clinical-assistant-agent.vercel.app · group capstone

- Agents **Report Analysis → Clinical Summary → Recommendation → Physician Gate** read lab-report PDFs and explain findings.
- **RAG over 8 clinical guidelines**, tested on 15 synthetic reports; serverless and **stores no patient data**.

Stack React · RAG · multi-agent orchestration · Vercel

HyperGlow Ordering App — DEPLOYED

Client product · HyperGlow UK · Jun – Aug 2026

- Restaurant self-service ordering with **behaviour-driven upsell** based on ordering patterns.
- Full stack — **Next.js, PostgreSQL, React Native** — in Docker on Linux, plus an **AI image-enhancement app** for food photos.

Stack Next.js · React Native · PostgreSQL · Docker · Python

Multilingual RAG System — GITHUB

Bangla + English question answering · github.com/shaif143

- RAG over a Bengali textbook: **PyMuPDF** extraction, **sentence-based chunking**, **MiniLM** embeddings and **FAISS** retrieval feeding a local LLM.

Stack Python · sentence-transformers · FAISS · PyMuPDF

Factory Instrument Detection
YOLO model detects instruments carried by factory staff to grant access.
Python · YOLO · OpenCV

Human Activity Recognition
Deep learning on **smartphone sensor time-series**.
Python · CNN · LSTM

Term Deposit Prediction
Bayesian-tuned classifier predicting bank term-deposit sign-ups.
Python · Scikit-learn · GitHub

Heart Disease Prediction
Classification on clinical features with EDA and model comparison.
Python · Scikit-learn · GitHub

Emergency Helpline System
Full-stack 999-helpline alternative for police and fire services.
NestJS · Next.js · REST API

Dhaka Guide
Console information system for places and transport in Dhaka.
University project · console app

06 ACHIEVEMENTS & LEADERSHIP

- **Emerging Leader Award** — DMH Foundation, 2024.
- **Dean's Honourable Mention List, twice** — AIUB.
- **Research mentoring** — three AIRIL research teams.
- **Team Leader, AIESEC Bangladesh** — led a team of four.
- **Presentations** to technical & non-technical audiences.
- **Positive ROI** on AI projects for several companies.

08 VOLUNTEERING & LANGUAGES

- **Volunteer IT Support**, Sholoana & Ashia Foundations — IT systems, volunteer mentoring, medical campaigns.
- **Languages:** Bangla (native), English (proficient), Hindi (conversational).
- **Interests:** photography, poetry, HCI, mentoring.

07 RESEARCH PUBLICATIONS

01 Advancing Monkeypox Diagnosis: A Novel Approach Using Custom Neural Networks.

02 DeepResVit: A Hybrid Deep Learning Approach for Ovarian Cancer Classification with XAI.

03 An Explainable Hybrid CNN-ANN Model for Smartphone-Based Human Activity Recognition in Healthcare Analytics.

04 AeroLung AI: An Explainable Machine Learning Framework for Lung Cancer Prediction in Smart Healthcare Systems.

05 Interpretable Machine Learning Techniques for Human Activity Recognition using Transfer Learning.

06 Improving Liver Disease Diagnosis through Ensemble Machine Learning Techniques.

09 REFERENCES

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See my agents running live.

Scan to try **ClinicalOKF** and **ClinicalAI**. The demos are protected — ask me for the secret access code.

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